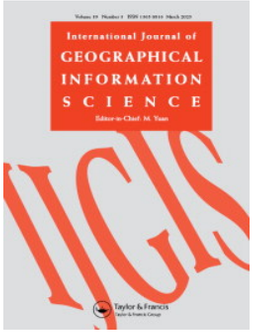

International Journal of Geographical Information Science

ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/tgis20

GeoTool-GPT: a trainable method for facilitating Large Language Models to master GIS tools

Cheng Wei, Yifan Zhang, Xinru Zhao, Ziyi Zeng, Zhiyun Wang, Jianfeng Lin, Qingfeng Guan & Wenhao Yu

To cite this article: Cheng Wei, Yifan Zhang, Xinru Zhao, Ziyi Zeng, Zhiyun Wang, Jianfeng Lin, Qingfeng Guan & Wenhao Yu (11 Dec 2024): GeoTool-GPT: a trainable method for facilitating Large Language Models to master GIS tools, International Journal of Geographical Information Science, DOI: 10.1080/13658816.2024.2438937

To link to this article: https://doi.org/10.1080/13658816.2024.2438937

Published online: 11 Dec 2024.

Submit your article to this journal

Article views: 580

View related articles

View Crossmark data

Citing articles: 2 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tgis20

INTERNATIONAL JOURNAL OF GEOGRAPHICAL INFORMATION SCIENCE
https://doi.org/10.1080/13658816.2024.2438937

Taylor & Francis
Taylor & Francis Group

RESEARCH ARTICLE

Check for updates

GeoTool-GPT: a trainable method for facilitating Large Language Models to master GIS tools

Cheng Wei[a*], Yifan Zhang[a*], Xinru Zhao[a], Ziyi Zeng[a], Zhiyun Wang[a], Jianfeng Lin[b], Qingfeng Guan[a,c] and Wenhao Yu[a,c]

[a]School of Geography and Information Engineering, China University of Geosciences, Wuhan, China; [b]Meituan, Beijing, China; [c]National Engineering Research Center for Geographic Information System, China University of Geosciences, Wuhan, China

ABSTRACT

Large Language Models (LLMs) excel in natural language-relevant tasks like text generation and question answering Q&A. To further expand their application, efforts focus on enabling LLMs to utilize real-world tools. However, their tool-use ability in professional GIS remains under explored due to two main challenges. Firstly, LLMs are usually trained on general-domain corpora, lacking sufficient and comprehensive GIS-specific data to align with professional knowledge, including understanding the functions of GIS tools. Secondly, researchers often need to combine multiple GIS tools to solve geospatial tasks. To address these challenges, we propose a trainable method to enable LLMs to master GIS tools. We curated a comprehensive set of resources: instruction-response data (GeoTool, 1950 instructions) to enhance the understanding of LLMs for GIS tools, instruction-solution data (GeoSolution, 3645 instructions) to improve their ability to generate tool-use solutions for geospatial tasks, and annotated instruction-solution evaluation data (GeoTask, 300 instructions) for evaluating LLMs' GIS tool-use proficiency. Using the collected training data (GeoTool and GeoSolution), we fine-tuned a professional-domain LLM called GeoTool-GPT based on an open-source general-domain LLM, the LLaMA-2-7b model. The experiment based on evaluation data validates our method's effectiveness in enhancing the tool-use ability of general-domain LLMs in the professional GIS domain, with the performance of our model closely approaching that of GPT-4.

ARTICLE HISTORY
Received 28 March 2024
Accepted 2 December 2024

KEYWORDS
Large language model; tool-use ability; instruction tuning; agent

1. Introduction

Large Language Models (LLMs) have demonstrated significant capabilities and immense potential within the field of natural language processing (Almarie and and Teixeira 2023, Hadi *et al*. 2023, Thirunavukarasu *et al*. 2023, Teubner *et al*. 2023). They exhibit remarkable creativity and linguistic proficiency, particularly in tasks such as knowledge question answering and article writing (Franceschelli and Musolesi 2023).

CONTACT Wenhao Yu yuwh@cug.edu.cn
*Authors with equal contribution to this paper.
© 2024 Informa UK Limited, trading as Taylor & Francis Group

Since the release of ChatGPT (OpenAI 2023a), there has been a surge of interest in LLMs from researchers across various disciplines, including computer science (Zhang *et al*. 2023), medicine (Karabacak and Margetis 2023), law (Yue *et al*. 2023), and GIS (Hochmair and Guan 2023, Ji and Gao 2023, Li and Ning 2023, Tao and Xu 2023, Zhang *et al*. 2024a, 2024c).

Some researchers focus on leveraging the powerful capabilities of LLMs, including extensive knowledge, semantic comprehension, reasoning, and planning, to assist in their tasks. For example, in the field of recommender systems, Gao *et al*. (2023) used ChatGPT to analyze user preferences by inputting user profiles and historical interactions. This approach can enhance the interactivity and interpretability of the recommendation process. In the GIS domain, researchers have also explored the effectiveness of LLMs in solving geospatial tasks. For instance, Mai *et al*. (2023) applied LLMs to tasks such as toponym recognition, location description recognition, and US state-level/county-level dementia time series forecasting, by inputting corresponding text into the models. Hu *et al*. (2023) proposed a geo-knowledge-guided GPT model for extracting location descriptions that go beyond simple place names from social media textual messages by using training examples as prompt-based guidance.

Given the strong performance of LLMs in these studies, some researchers are exploring expanding their applicability to various scenarios by integrating real-world tools. Unlike depending solely on the inherent knowledge of LLMs, incorporating external tools enables them to interact with additional resources, such as data and the latest information. Some researchers have validated the effectiveness of combining LLMs with tools to address complex tasks (Dai *et al*. 2023, Xi *et al*. 2023). For instance, the fusion of LLMs with professional GIS tools has proven valuable for solving geospatial tasks and assisting mapping (Zhang *et al*. 2024a, 2024b), while combining LLMs with professional chemistry tools can facilitate tasks in chemistry such as organic synthesis and drug discovery (Bran *et al*. 2023). However, despite the demonstrated value of incorporating tools into LLM workflows, these studies often rely on state-of-the-art (SOTA) LLMs, such as the closed-source models ChatGPT and GPT-4, which have opaque inner mechanisms (Qin *et al*. 2023). Although open-source LLMs have demonstrated robust performance in natural language-related tasks, they still exhibit limited tool-use capabilities (Tang *et al*. 2023).

Some researchers also recognized this problem, and proposed strategies to enhance the tool-use capabilities of open-sourced LLMs (Shen *et al*. 2024). For example, Qin *et al*. (2023) introduced ToolLLM, a framework designed to improve the general tool-use proficiency of the open-source LLM, LLaMA. This framework focus on general scenarios and included an instruction-tuning dataset to fine-tune LLaMA's tool-use abilities. However, while there has been significant attention on enhancing tool-use abilities in general domains, the applicability of open-source LLMs in the professional GIS domain is still being explored, primarily due to two main challenges. Firstly, LLMs are typically trained on large-scale corpora from general domains. The lack of sufficient and comprehensive training data from the GIS domain hinders their alignment with professional knowledge, including understanding the functions of GIS tools. Secondly, the GIS functionality is difficult to identify and use without specific

training of GIS skills (Gao and Goodchild 2013). Reducing the burden on users, particularly novice users, to know the exact commands of operations and their sequences to perform for achieving the goal they want to achieve is essential (Zhu *et al.* 2021). Furthermore, traditional approaches to solving GIS tasks require manual intervention, primarily involving the application of domain expertise in combination with various GIS tools to address geospatial problems. For example, using the ArcGIS's ModelBuilder module to construct geoprocessing workflows for specific GIS issues requires careful design by professionals and is typically tailored to specific tasks, making it difficult to generalize or extend to other applications.

An illustrative example is provided in Figure 1. In Figure 1a, a general-domain LLM can easily answer the question presented if corresponding knowledge or information is available in the training data. However, for the question posed in Figure 1b, geographical information about the 'shortest distance' between two places may be lacking in the training data, rendering the LLM unable to respond. In practice, researchers

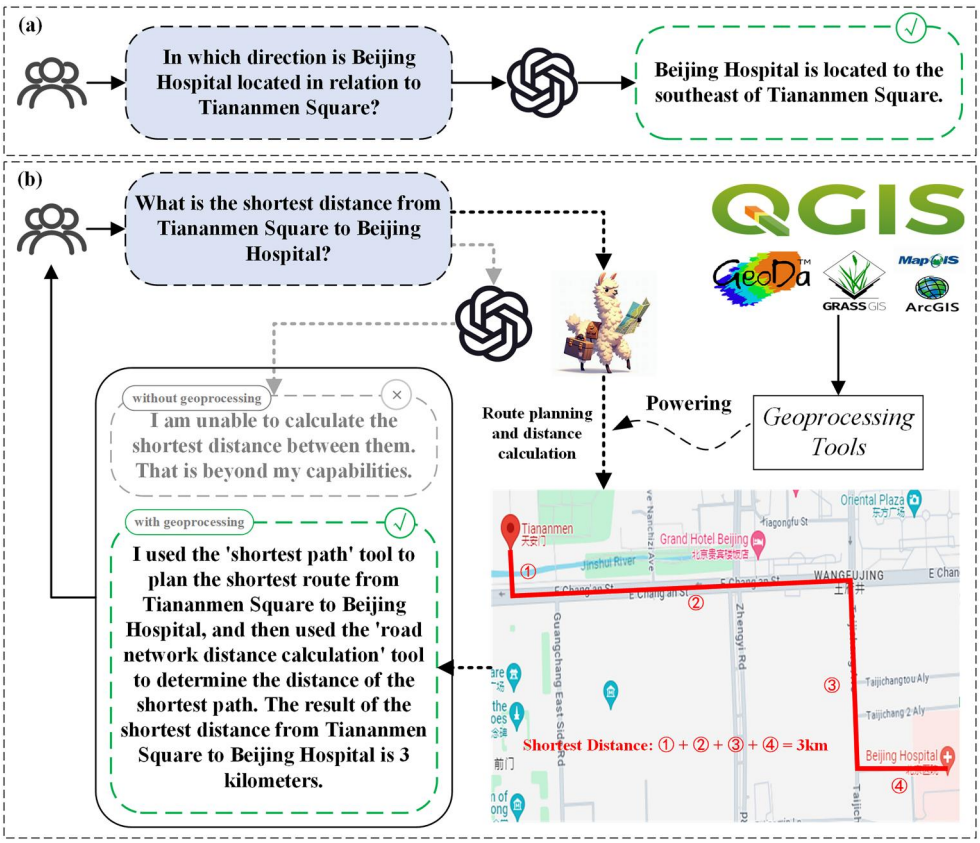

Figure 1. The challenge lies in enabling LLMs to utilize GIS tools for solving geospatial tasks. While general-domain LLMs may easily answer questions if corresponding knowledge or information is available in the training data (a), geospatial tasks often require multiple GIS tools. Direct geographical information may be lacking in the training data, rendering the LLM unable to respond. Thus, enabling LLMs to understand the function of each GIS tool and determine the appropriate sequence of tools for geospatial tasks is critical (b).

often combine multiple GIS tools to address such challenges. For instance, they may use the 'shortest path' tool to plan a route and the 'road network distance calculation' tool to calculate the distance of the route. Thus, enabling LLMs, as illustrated in Figure 1b, to understand the function of each GIS tool and determine the appropriate sequence of tools for a specific geospatial task presents challenges.

To address these challenges, we propose a trainable approach designed for facilitating LLMs to master GIS tools. In general, fine-tuning LLMs with professional training data is an effective way to align them with professional knowledge and improve their ability to follow instructions. According to the method, we curated a comprehensive set of resources: instruction-response data (GeoTool, 1950 instructions) to enhance the understanding of LLMs for GIS tools, instruction-solution data (GeoSolution, 3645 instructions) to improve their ability to generate tool-use solutions for geospatial tasks. Since manually collecting this data is time and resource-consuming, we employ a self-instruct strategy. This involves generating instruction data using a robust LLM (GPT-4). Using the collected training data (GeoTool and GeoSolution), we fine-tuned a professional-domain LLM, GeoTool-GPT, based on an open-source general-domain LLM, the LLaMA-2-7b model (Touvron *et al*. 2023). Finally, to evaluate the tool-use proficiency of LLMs in geospatial tasks, we also collected annotated instruction-solution evaluation data (GeoTask, 300 instructions). The experiment based on evaluation data shows that the fine-tuned GeoTool-GPT performs exceptionally well in GIS spatial analysis tasks, approaching the state-of-the-art GPT-4. This result validates the effectiveness of our method in enhancing the tool-use ability of LLMs in the professional GIS domain. Our model can serve as the basis for integration with GIS tools, enabling the deployment of intelligent applications capable of autonomously solving GIS spatial analysis tasks. In particular, our paper makes the following contributions:

- Firstly, we introduced an open enhanced GIS-specific large language model named GeoTool-GPT, designed to align with human intentions of using GIS tools to solve geospatial tasks. GeoTool-GPT can serve as the core module of an AI Agent, perceiving GIS task requirements through natural language and making decisions regarding the use of GIS tools. These core solutions can then implemented by the AI Agent, which is integrated with real-world tools, to achieve intelligent resolution of GIS tasks. GeoTool-GPT shifts the traditional pattern of GIS task resolution, which has been dominated by professionals, by leveraging its combined capabilities of intelligent demand understanding and smart tool scheduling, making the resolution of GIS tasks more convenient and user-friendly.
- Secondly, we proposed a systematic approach for constructing the data required to train and evaluate GeoTool-GPT model. This method primarily involves carefully designing specialized system prompts to guide GPT-4 in self-generating instruction-following training datasets and employing professionals to annotate evaluation datasets. Based on this approach, we curated a comprehensive set of resources, including instruction-response data (GeoTool), instruction-solution data (GeoSolution), and annotated instruction-solution evaluation data (GeoTask). Our empirical study validates the effectiveness of using the generated GIS instruction-tuning data.

2. Related work

Large Language Models (LLMs) have taken the initiative in the field of natural language processing, achieving remarkable performance across various text-based tasks and demonstrating powerful capabilities in semantic understanding, reasoning, and planning (Sun *et al*. 2021, Jin and Zhuo 2022, Chang *et al*. 2023). Recently, LLMs research in the multimodal domain has rapidly advanced, yielding significant achievements, particularly in the field of computer vision, where a series of end-to-end visual-language models have been explored for various research topics. For example, tasks such as vision-language navigation (Hao *et al*. 2020) and Habitat (Andrew *et al*. 2021) necessitate the embodied AI agent to adhere to natural language instructions and execute a series of actions to achieve objectives within visual environments. In the domain of image editing, InstructPix2Pix (Tim *et al*. 2022), for instance, operates by modifying images based on human instructions, where an input image and written instructions guide the agent in the editing process. In other studies, efforts are made to extend LLMs to accommodate multimodal data via specific framework like LangChain, which is a framework for developing applications powered by language models. Examples include Visual ChatGPT (Wu *et al*. 2023), X-GPT (Zou *et al*. 2022), MM-REACT (Yang *et al*. 2023), VisProg (Tanmay and Aniruddha 2022), and ViperGPT (Dídac *et al*. 2023). The methods proposed in previous studies were then adapted for the remote sensing (RS) domain, since remote sensing images share similar data structure with natural images. For instance, SkyEyeGPT (Zhan *et al*. 2024) addresses RS visual language understanding tasks by injecting visual features of remote sensing images into the language domain using an alignment layer. RS-ChatGPT (Guo *et al*. 2024) leverages ChatGPT to connect various remote sensing models and constructs an agent for solving remote sensing image interpretation tasks through designing visual prompts.

Integrating artificial intelligence methods to solve GIS tasks has been focused by many researchers (Li 2020, Janowicz *et al*. 2020, Mai *et al*. 2022). Nevertheless, there has been little research focused on how LLMs handle GIS tasks and adapt to the spatial data involved. Integrating a LLM with a professional agent framework may present a promising approach, where the LLM is responsible for understanding GIS task requirements and providing task solutions, while the agent leverages the generated solution for interacting with data by GIS tools. For example, Zhang *et al*. (2024a) utilized the LangChain framework to build an AI agent named GeoGPT, capable of handling geospatial tasks by leveraging models like GPT-4 and GPT-3.5 provided by OpenAI through commercial interfaces. By harnessing the powerful requirement comprehension capabilities of the GPT-4 model and embedding customized GIS tools within the system prompts, the AI agent can autonomously invoke GIS tools to solve spatial tasks. Additionally, Li *et al*. (2023) also capitalized on GPT-4's advanced capabilities, including requirement understanding and code generation, by designing system prompts that enable GPT-4 to interpret task requirements and generate code to complete the tasks. Nevertheless, intelligent strategies for addressing GIS-related problems primarily rely on the robust intrinsic capabilities of large parameter models, such as the trillion-parameter GPT-4 (OpenAI 2023b). Therefore, as highlighted in the discussion sections of relevant studies, it is crucial to develop specialized and accessible

LLMs capable of following instructions related to GIS tasks. Moreover, the approach of placing all available GIS tools within system prompts to improve the model's mastery of tools, as done in the GeoGPT framework during the construction of the AI agent, is not a permanent solution. Constrained by the limited context window of LLMs, this framework exhibits poor scalability for handling extended tasks. It would be more effective to directly train a specialized large model for GIS tools, embedding the GIS tools within the model's parameters.

For developing professional LLMs in GIS, the instruction-following data for training is crucial, for no readily available data for GIS task-solving. In the literature, the synthetic-data-generation strategy based on LLMs is widely recognized. For instance, Wang *et al*. (2022) introduced a Self-Instruct framework aimed at constructing an instruction dataset to enhance the instruction-following capabilities of LLMs, which was achieved through bootstrapping LLMs with abundant knowledge to generate content. Subsequently, Taori *et al*. (2023) implemented the Self-Instruct framework by utilizing the Text-Davinci-003 model to generate 52,000 instructions for model fine-tuning. After being fine-tuned on the generated instructions, LLaMA, a model of 7 billion parameters, presents comparable performance with Text-Davinci-003. More studies further corroborated the effectiveness of datasets by the self-instruct strategy (Ding *et al*. 2023, Chiang *et al*. 2023, Xu *et al*. 2023).

Upon previous contributions, researchers in various domains are exploring the development of LLMs tailored to their specific fields to offer more specialized services. For instance, Yue *et al*. (2023) introduced an intelligent legal system named DISC-LawLLM that employs a law-specific LLM fine-tuned for the legal domain. This model is able to offer a comprehensive array of legal services. Deng *et al*. (2023) presented a fundamental language model designed for the comprehension and utilization of geoscience knowledge. In the aspect of utilizing tools to address tasks, Qin *et al*. (2023) fine-tuned LLaMA for using web APIs to fulfill generic task instructions. Despite the thriving research on fine-tuning open-source LLMs, there has been a lack of exploration in instruction-following for GIS tasks. We aim to fill this gap by introducing an open large GIS-task-specific model that demonstrates proficiency in instruction-following. Furthermore, the model leverages the instruction-following data generated through an improved Self-Instruct strategy for training GIS LLMs.

3. Generating instruction-following data for enhancing GIS tool-use ability of LLMs

Instruction-following refers to the ability of LLMs to execute specific operations based on given instructions or task descriptions. For example, if the model is instructed to generate a piece of elegant prose describing a sunrise, it must understand the directive and complete the task by generating the requested text. This capability is particularly crucial for LLMs because it directly impacts how the model interprets and responds to user needs. The self-instruct strategy is a commonly used method for generating instruction-following samples, aiming to improve the model's task execution abilities through self-generated instructions. This approach involves two main steps: instruction generation and task execution. In the first step, the LLMs generate a task

instruction within a specific domain based on a designed prompt, while in the second step, the LLMs execute the task as per the generated instruction, thereby producing a set of instruction-following data. Unlike generation of instruction-following data for general tasks, generating instruction-following data for using GIS tools requires initial data, i.e. metadata of text description for GIS tools. This enables LLMs to understand the toolkit available for solving GIS tasks, and allows LLMs to better self-instruct in generating instructions relevant to usage of GIS tools. Therefore, we propose a systematic pipeline for constructing instruction-following data specific to GIS tasks, as shown in Figure 2.

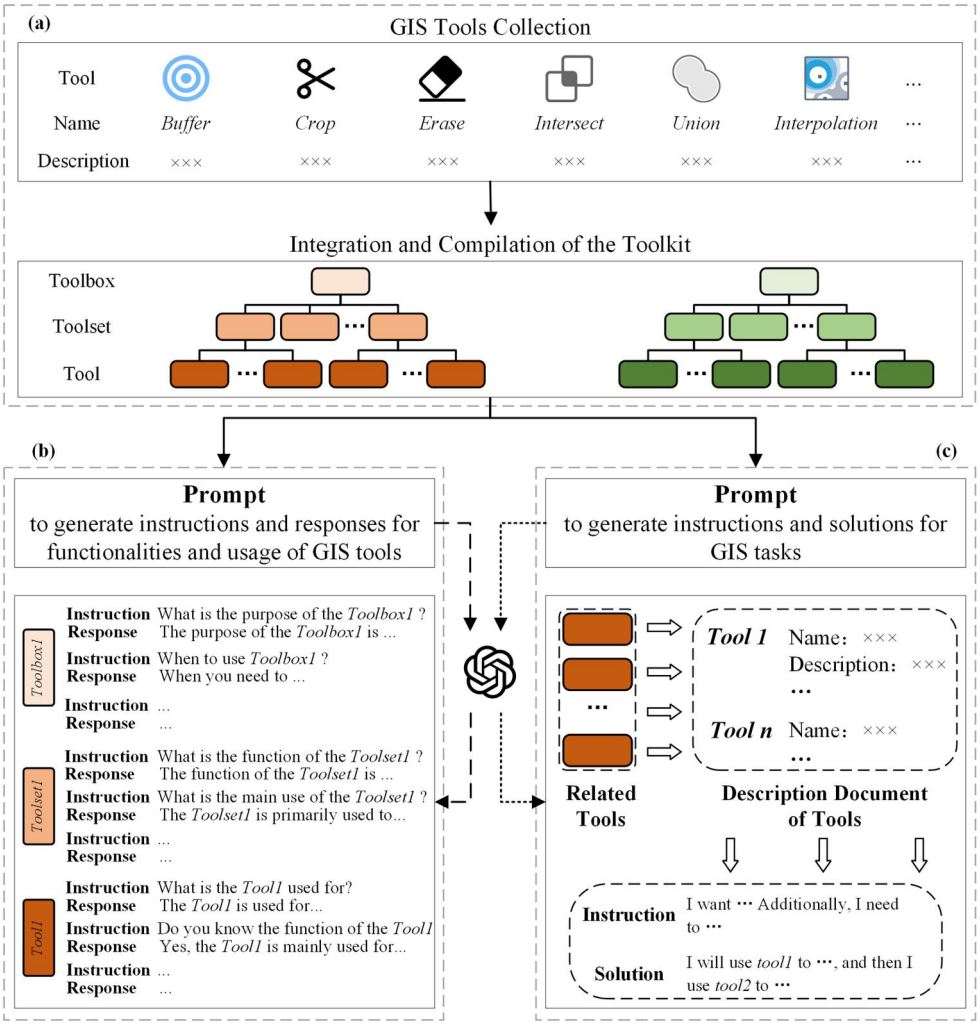

Figure 2. A systematic pipeline for constructing instruction-following data specific to GIS tasks. (a) Collection, integration and compilation of GIS tools. (b) Self-instructing GPT-4 to generate instruction-response samples (GeoTool) for LLMs to understand the functionalities and the usage of GIS tools. (c) Self-instructing GPT-4 to generate instruction-solution samples (GeoSolution) for enhancing the ability of LLMs on using GIS tools.

- Initially, we curate commonly used tools related to GIS operations to form a geoprocessing toolkit, which is then integrated and organized in a hierarchical structure (Figure 2a).
- Subsequently, leveraging the capabilities of GPT-4 (OpenAI 2023b), we generate an instruction-response sample set, which encompasses various conceptual levels of GIS tools through crafted prompts. This process aims to facilitate the understanding of the conceptual differences and practical applications of GIS tools for LLMs (Figure 2b).
- Next, within the integrated toolkit, we direct GPT-4 to explore potential combinations of relevant tools that could be covered in a natural way when executing a single instruction. Following this exploration, we guide GPT-4 in generating instructions and solutions based on the documentation of the pertinent tools (Figure 2c).

3.1. Hierarchical compilation of geoprocessing toolkit

Our objective is to instruct open-source LLMs to comprehend the functionality and application contexts of GIS tools, with a focus on mastering the sequential usage of multiple interconnected tools for specific geospatial tasks. Therefore, the basic task during the tools compilation process involves gathering documentation for these tools, primarily focusing on collecting textual descriptions rather than the tools' procedural implementations. We meticulously standardize 172 comprehensive documentations for fundamental GIS tools, inclusive of two essential fields, i.e. names and descriptions of tools. In addition, since we aim to use a relatively large number of tools, a hierarchical organization of GIS tools is developed to enhance the ability of LLM in invoking tools from the geoprocessing toolkit accurately. Specifically, we have organized all the tools in the geoprocessing toolkit into two main toolboxes, each of which is further divided into several toolsets. Each toolset contains a number of tools that perform similar or related functions. Meanwhile, we enrich both the toolboxes and toolsets with linguistically precise names and descriptions to help understand their functionalities. We provide a comprehensive overview of the integration and compilation of GIS tools in ***Appendix A***.

3.2. GeoTool: instruction-response samples for enhancing LLM's understanding of functionalities and usage of GIS tools

Previous studies have explored the effectiveness of integrating LLMs with tools to address related tasks (Bran *et al*. 2023, Wu *et al*. 2023). These approaches typically include tool documentation, comprising names and descriptions, either after the task instructions or within the system prompts, with the aim of informing the LLM about task requirements and providing a diverse set of tools for use. However, as the number of tools increases, the inherent token limit within the model's context window imposes an additional burden on the LLM's contextual understanding, thereby reducing its effectiveness in handling tasks (Li *et al*. 2022b, Rae *et al*. 2021). In this way, a better way is to directly inject professional knowledge about GIS tools into the LLM.

This configuration enables the fine-tuned language model to emulate the expert of GIS who possesses comprehensive knowledge about the functionalities and usage of GIS tools before tackling GIS requirements.

Therefore, we propose the generation of instruction-response samples based on the self-instruction of GPT-4. In our method, the instructions revolve around the functionalities and usage scenarios of tools, with corresponding responses being contextually appropriate replies based on the tool descriptions. We have meticulously designed the prompt to instruct GPT-4 in generating instructions and responses involving hierarchical organization of toolboxes, toolsets and tools, as shown in Figure 3. To enhance GPT-4's understanding of our requirements regarding the generated instructions, we have leveraged its exceptional few-shot capabilities by incorporating several straightforward examples within the prompts. Subsequently, we guide GPT-4 to generate responses based on the tool documentation while allowing for the occasional creation of simple usage examples to aid comprehension. Simultaneously, we have standardized the output of GPT-4 to facilitate data extraction and aggregation. Based on the prompt designed for GPT-4, a total of 1950 instruction-response pairs are ultimately yielded in GeoTool to help model for learning the usage of the collected tools.

3.3. GeoSolution: instruction-solution samples for enhancing the GIS tool-use ability of LLM

In order to enhance the tool-use efficacy of LLM in addressing GIS tasks, another emphasis lies in generating task instructions and solutions for GIS tasks. Specifically, we direct GPT-4 to understand the functionalities of GIS tools by randomly sampling several toolsets from each toolbox at a time and feeding the tool names and descriptions into GPT-4. And subsequently, we employ a reverse generation approach to produce GIS task instructions incorporating the usage of multiple tools. However, randomly sampling multiple tools from the complete tool set could yield ineffective results, as the sampled tools may lack functional and operational coherence. This deficiency results in a lack of natural sequencing for their invocation when completing an instruction, and that could lead to a sparse and less effective generation of instructions.

To mitigate this dilemma, we instruct GPT-4 to explore the GIS tool set based on tool descriptions and inherent GIS knowledge, to seek a subset of tools that may naturally coalesce within a single instruction. As presented in Figure 4, specific prompts are designed to instruct GPT-4 to generate GIS task instructions from descriptions of the post-exploration GIS tools. The prompt incorporates few-shot examples to align instruction generation behavior of GPT-4 with our intended objectives. Subsequently, we task GPT-4 with presenting the calling sequence of the related tools, which is capable of executing the instruction of GIS task. This is accompanied by a detailed explanation of each tool's usage and rationale for their sequential application. The explanation essentially plays the role of a Chain of Thought (CoT) (Wei *et al*. 2022) in the solution. We advocate for comprehensive explanation as a supplementary solution, considering that providing a mere sequence of tool calls may prove insufficient.

You will get an entity among toolbox, toolset, and tool, each containing a corresponding name and description. These toolboxes/toolsets/tools are professional tools in the field of geographic information processing. Imagine yourself as a domain expert with this background knowledge, and I will provide you with a geoprocessing tool and its corresponding description. Your task is to produce 10 related queries and their responses based on your understanding of the tool name and description of usage to illustrate the possible usage scenarios or methods of the tool from different perspectives.

For example, you are provided with a tool called Buffer, and you can generate queries like this:
Query 1: Do you know the Buffer tool in the GIS field?
Query 2: Are you familiar with the Buffer tool in GIS for creating geographic areas?
Query 3: What is the Buffer tool used in the GIS field?
...

It can be found that similar queries are mainly based on tool concepts. You will need to use your imagination to generate similar queries from different perspectives to elicit responses to the conceptual and practical uses of GIS tools.

These queries should display a diverse range of sentence structures: some queries should be in the form of imperative sentences, others declarative, and yet others interrogative. Equally, they should encompass a variety of tones, with some being polite, others straightforward.

After that, what you need to do is to give a corresponding response to each query based on the provided tool description and your own understanding. Responses may occasionally include simple examples to aid understanding.

Queries and responses need to be matched one by one.

Use the following template to deliver your final answer:

Query1 : ... & response1 : ...
Query2 : ... & response2 : ...
Query3 : ... & response3 : ...
...
--- (a)

The provided tool is: "**Erase**", whose description is "Creates a feature class by overlaying the Input Features with the polygons of the Erase Features. Only those portions of the input features falling outside the erase features outside boundaries are copied to the output feature class.". (b)

Query : Could you explain what the Erase tool does in GIS? & response : Certainly, the Erase tool in GIS is primarily used for overlaying the input features with the polygons of the Erase features. It then creates a new feature class where only those parts of the input features that do not intersect with the Erase features are included. (c)

Figure 3. (a) The system prompt used for instructing GPT-4 in generating GeoTool (instructions and corresponding responses) for enhancing LLM's understanding of functionalities and usage of GIS tools; (b) the actual input prompt combined with the system prompt to form the complete prompt; (c) an example of the sample data generated based on the complete prompt.

Additionally, given that the generated task instructions and solutions involving tool invocation sequences and detailed explanations occasionally exhibit a tendency towards extended length, we restrict GPT-4 to generate only one instruction-solution pair per iteration to prevent the generation of a volume of text that could surpass the limitations imposed by the output token constraints. This can also mitigate the likelihood of encountering output disarray or hallucination. Finally, a total of 3645 instruction-solution pairs are collected in GeoSolution, meticulously curated to refine the adeptness of models in utilizing GIS tools.

You will be provided with several toolsets in a toolbox, toolset descriptions, all of each toolset's available tool functions (apis), the descriptions of these tool functions. You need to first know the name of each tool and a description of its function, and then think about possible related tool combinations in the same toolset or different toolsets. Related tools are tools that can be used for a given query; those related tools have to strictly come from the provided tool names. Your task involves creating one varied, innovative, and detailed user query where the related tools in one or more toolsets you think out previously can be used to resolve the query exactly!

For instance, given three related GIS tools in three toolsets 'Extract', 'Proximity', and 'Statistics': the tool of 'Select' in 'Extract', the tool of 'Near' in 'Proximity' and the tool of 'Summary Statistics' in 'Statistics' may have usage correlations in their function descriptions. Your query should articulate something akin to: "We're devising a traffic management plan for New York City. We need to identify all roads within each district, calculate the distance between each road and the nearest hospital, and generate statistics on these distances. "
Another example, given two related GIS tools in one toolset 'Extract': 'Split' and 'Select'. Your query should articulate something akin to: "For a study on bird migration, I want to split a global map into various regions based on bird species."

These queries exemplify how to utilize tool calls of all the given related tools. A query that uses tool calls of only one tool will not be accepted. Additionally, generate random information for parameters such as IP address, location, coordinates, etc. For instance, don't merely say 'an address', provide the exact road and district names. Don't just mention 'a product', specify wearables, milk, a blue blanket, a pan, etc. Don't refer to 'my company', invent a company name instead.

For each query, there should be multiple related tools; for different queries, overlap of related tools should be as little as possible. Keep in mind that for each query, invoking just one tool won't suffice; each query should call at least two tools. However, try to avoid explicitly specifying which tool to employ in the query. Each query should consist of a minimum of thirty words.

After that, you should explicitly list the order in which these related tools are called to effectively address the respective query. Following this, provide an explanation for why you believe the specified tool order effectively resolves the query. For example, you can say, 'I use the ××× tool to ...'!!!

Deliver your response in this format:

Query: ..... & related tools:[(toolset1, tool3), (toolset2, tool1), (toolset4, tool1)...] & tool call sequence: (toolset2, tool1)->(toolset1, tool1)->(toolset1, tool3)->... & explanation: I use the (toolset2, tool1) to ... Then I will use (toolset1, tool1) ...

(a)

The provided toolsets in a toolbox are blow: {
"name": "Overlay", "description": "...",
"tool_list": [{
"name": "Erase", "description":"...";
"name": "Intersect", "description":"...";
...
};
"name": "Proximity", "description": "...",
"tool_list": [{
"name": "Buffer", "description":"...";
"name": "Near", "description":"...";
...
};
...
}, which belongs to the toolbox of "Analysis".
(b)

Query: We're planning a community event in the park in downtown San Francisco. We need to estimate how many people from the nearby residential areas might attend. I need to know the population density around the park within a 1-mile radius. & related tools: [('Proximity', 'Buffer'), ('Overlay', 'Intersect'), ('Statistics', 'Summary Statistics')] & tool call sequence: ('Proximity', 'Buffer')->('Overlay', 'Intersect')->('Statistics', 'Summary Statistics') & explanation: I first create a 1-mile buffer around the park using the 'Buffer' tool. The 'Intersect' tool helps me overlay this buffer with the population data. I then use 'Summary Statistics' to calculate the population density within this buffer.
(c)

Figure 4. (a) The system prompt used for instructing GPT-4 in generating GeoSolution (instructions and corresponding solutions) for enhancing GIS tool-use ability of LLMs; (b) the actual input prompt combined with the system prompt to form the complete prompt; (c) an example of the sample data generated based on the complete prompt.

4. Augmented Large Language Model for GIS tool use

The endeavor of training a LLM tailored for addressing GIS tasks from scratch incurs substantial costs, necessitating extensive computational and temporal resources, alongside a substantial foundational training corpora (Roberts *et al*. 2023, Rogers *et al*. 2023). In contrast, the pragmatic approach of fine-tuning pre-existing and pre-trained LLMs presents an optimal solution for expediently introducing such models to specialized domains. Therefore, we select the open-source LLaMA-2-7b, a model with 7 billion parameters as the base model for fine-tuning. We opted for LLaMA-2-7b primarily due to its demonstrated capacity to generate responses that rival those of closed-source models like ChatGPT and PaLM across diverse conversation benchmarks, despite having only 7 billion parameters.

Considering the efficiency for fine-tuning a LLM, we embrace a lightweight fine-tuning strategy LoRA (Hu *et al*. 2021). Without modifying the parameters of the pre-trained model, LoRA injects two low-rank matrices into initial LLMs for training, facilitating model customization and adaptation. This approach significantly diminishes training costs and enhances the performance of our model. Building upon this methodology, we introduce GeoTool-GPT, an augmented large language model finely tuned on two RTX 4090 GPUs using the LoRA strategy. GeoTool-GPT is developed on the instruction-response and instruction-solution sample sets as presented in Sections 3.2 and 3.3. After the fine-tuning process, our model can offer specialized GIS task instruction-solving strategies.

5. Evaluation and case studies

5.1. Evaluation framework

There is currently a lack of an efficient evaluation framework and benchmark dataset for assessing the GIS tool-use proficiency of LLMs. To address this gap, we have constructed an evaluation benchmark. To ensure the effectiveness of the evaluation, all the instruction-solution samples are manually annotated and checked.

It needs to be emphasized that the instruction-solution generation strategy proposed in Section 3.3 is not applicable during the construction of the benchmark dataset. As shown in Figure 5a, during the instruction-solution generation process, GPT-4 explores related tools only within the sampled toolsets to reverse-generate the instructions. However, constructing the benchmark dataset should consider scenarios where a broader set of tools is available, ensuring that the evaluation results are convincing. Generally, in GIS, there are a large number of GIS tools that can be used to solve geospatial tasks. Given the extensive of GIS tool set, LLMs need to first understand the geospatial task requirements as well as the functionality of each GIS tool, then select the correct tools while providing their invocation sequences accurately. Constrained by the limitations of the context window tokens, even outstanding LLMs like GPT-4 cannot achieve ideal accuracy in generating solutions for GIS tasks by providing all the GIS tools as the input.

In light of this, as shown in Figure 5b, we leveraged the professional expertise of individuals to construct an evaluation instruction set that extensively covers tool usage

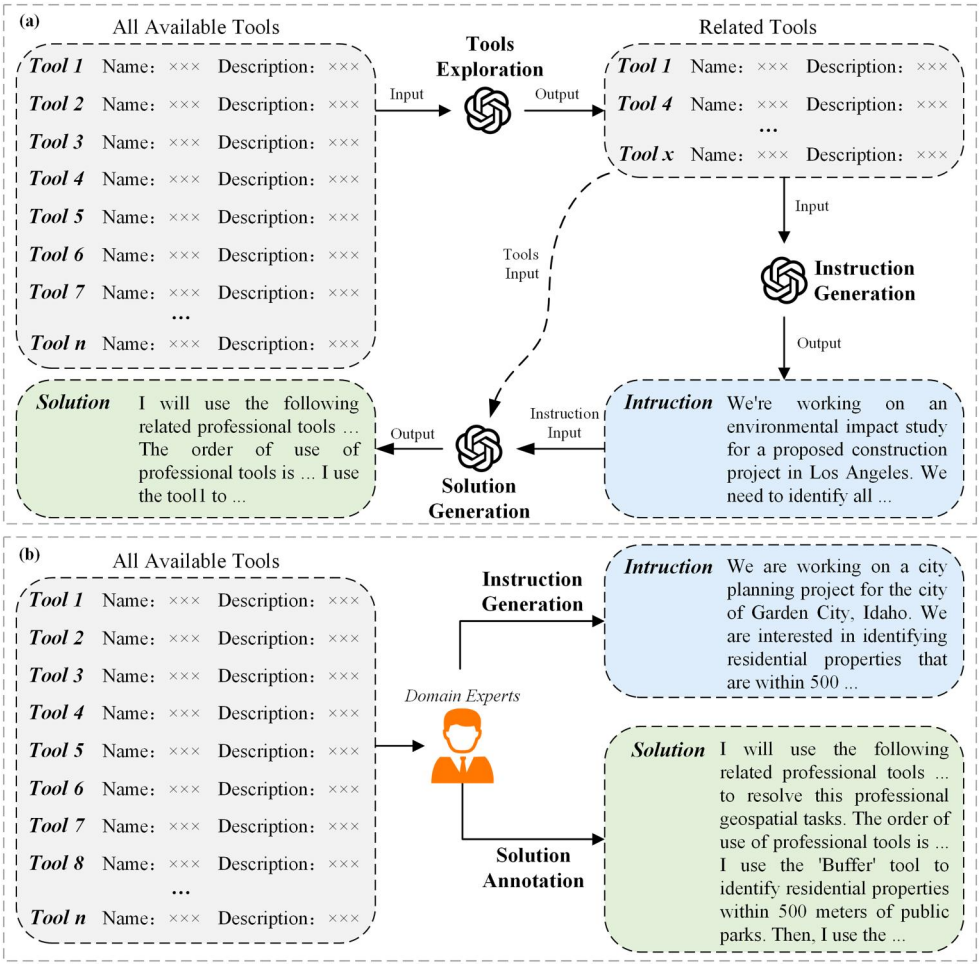

Figure 5. Instruction-solution samples construction strategies for distinct purposes. (a) Construction of instruction-solution samples intended for training LLMs primarily with the help of GPT-4 (GeoSolution). (b) Construction of instruction-solution samples intended for evaluating the tool-use ability of various LLMs (GeoTask).

in various scenes. The results follow the same style as the training samples in GeoSolution, and they are then provided to professionals for annotations. Finally, the manually constructed evaluation dataset GeoTask comprises 300 instruction-solution pairs related to GIS spatial analysis tasks. The first 100 pairs predominantly involve the use of GIS tools from the *Analysis* toolbox, while the subsequent 200 pairs predominantly involve the use of GIS tools from the *Spatial Analyst* toolbox.

Another focus in constructing the evaluation framework lies in conducting a cross-model comparison of results generated by baseline LLMs using the same benchmark dataset. However, owing to the extensiveness of GIS tools, the descriptive documentation for these tools may surpass the token limit of the context window in typical LLMs. For instance, the context window of 4096 tokens in GPT-3.5-turbo is inadequate to encompass all tool documentation, resulting in the incapacity of the model to fulfill instructions. Consequently, we selected the checkpoint of GPT-3.5-turbo-1106, which is

Image yourself as a geographic information science expert, and you will be provided with instructions for a geoprocessing task, as well as a range of geoprocessing tools in multiple toolsets. You first need to know the name of each tool and its corresponding tool usage description. After that, your task is to solve the requirements in the directive using the tools provided. The tools used may be from the same toolset or from different toolsets. You need to think about what tools you need to complete the tasks instructed. Then you need to output the names of the tools used to solve this geoprocessing task and the order in which they were called, and provide an explanation of the use of these tools and the order in which they were called.

For example, given the instructions for a geoprocessing task: "I am planning to build a wind farm at coordinates (50.2356, -120.6436). I need to find out if there are any inhabited areas within 5km and 10km from the intended location." After understanding the provided tool names and tool descriptions, you use the tools in this order: 'Multiple Ring Buffer' -> 'Generate Near Table'.

The provided toolsets are blow: {toolsets}

Deliver your response in this format:
---
tool call sequence: (toolset3, tool1)->(toolset2, tool1)->(toolset1, tool3)->.... & explanation: I use the (toolset3, tool1) to ...

Figure 6. The system prompt used for guiding the baseline large language models to better answer the instructions of GIS tasks.

characterized by a context window limit of 16,385 tokens, which presents comparative instruction-following capability compared to GPT-3.5-turbo. Then, we can implement the comparative analysis of performance with our GeoTool-GPT. We also selected the Gemini-Pro and GPT-4-0613 checkpoints as baseline models to compare the differences between our fine-tuned model and the state-of-the-art models.

Since the information of all available GIS tools has been injected into our GeoTool-GPT model during the training process based on the collected GeoTool and GeoSolution, we only need to input the instructions in GeoTask to the model during the evaluation process. However, to conduct an effective and fair comparison, a strategy is designed for general-domain baseline LLMs. Specifically, for the GPT-3.5-turbo-1106, Gemini-Pro, and GPT-4-0613 models, we designed a system prompt to provide them with the collected GIS tools and corresponding descriptions, as illustrated in Figure 6.

5.2. Evaluation

Upon completing the aforementioned procedures, we acquired the results from four models on GeoTask. All the results are evaluation by experts. Generally, when completing a GIS task instruction, the more tools involved often mean the more complex the GIS requirements are. To conduct a more thorough evaluation, based on the annotated solutions, we categorized the instructions according to the number of tools invoked to execute a single command. For each subset of instructions, we independently evaluated the outputs of different models and calculated the rates of *Accuracy*, defined as the ratio of the number of correctly solving instructions to the total number of instructions of the corresponding category. A solution is considered correct if the tools generated by the model, when executed as instructed, can successfully solve

Table 1. Comparative statistics of the performance of three general-domain large language models (GPT-3.5-turbo-1106, Gemini-Pro, and GPT-4-0613) and the proposed GeoTool-GPT in resolving GIS tasks.

		Number of correctly solving instructions (accuracy)			
The count of needed tools	Instruction count	GPT-3.5-turbo-1106	Gemini-Pro	GPT-4-0613	GeoTool-GPT
2	18	6 (33.3%)	6 (33.3%)	15 (83.3%)	15 (83.3%)
3	121	39 (32.2%)	39 (32.2%)	98 (81.0%)	86 (71.1%)
4	116	36 (31.0%)	29 (25.0%)	83 (71.6%)	71 (61.2%)
5	34	4 (11.8%)	5 (14.7%)	21 (61.8%)	19 (55.9%)
6	7	2 (28.6%)	4 (57.1%)	6 (85.7%)	5 (71.4%)
7	2	0 (0.0%)	0 (0.0%)	1 (50.0%)	1 (50.0%)
8	2	1 (50.0%)	1 (50.0%)	2 (100.0%)	1 (50.0%)
Total	300	88 (29.3%)	84 (28.0%)	226 (75.3%)	198 (66%)

the target task. This criterion accounts for the potential non-uniqueness of tool usage in solving the same task, as certain problems may have multiple valid solutions. For example, a spatial analysis task may be completed using either a single Tabulate Intersection tool or a combination of Spatial Join and Summary Statistics tools. However, if the generated tools are insufficient to complete the task, or if the proposed sequence of tools is logically or practically flawed (e.g. failing to adhere to required preprocessing steps for certain analyses), the solution is deemed incorrect. To ensure consistency, professionals reviewed the outputs to determine whether the solutions met these criteria. All the results are listed in Table 1.

From the results, it is evident that our GeoTool-GPT can proficiently comprehend both simple and complex GIS requirements, providing accurate solutions with high precision. Additionally, the performance of GeoTool-GPT notably exceeds that of GPT-3.5 as well as Gemini-Pro, and closely approaches that of GPT-4. The result validates the efficacy of our fine-tuned GeoTool-GPT model in addressing GIS task instructions, and also affirms the effectiveness of our collected instruction-following data for enhancing the tool-use ability of open-source general LLM.

5.3. Case studies

In this section, we present two case studies to demonstrate the performance of our GeoTool-GPT in efficiently addressing various geospatial tasks. Specifically, we chose instructions corresponding to different tool invocation chains from the GeoTask. This enables us to showcase the performance of GeoTool-GPT when dealing with different use requirements. Then, task-specific solutions generated by our model serve as workflows that can be readily implemented using modules in GIS platform such as ModelBuilder, enabling the efficient execution of geospatial tools. Additionally, these workflows can serve as a basis for constructing advanced AI agents in future work, allowing for seamless automation of tool parameter setting and execution within GIS software.

Firstly, we present a case concerning spatial overlay analysis, as illustrated in Figure 7. Specifically, the user's requirement is to identify suitable areas for a real estate development project. The primary criteria include proximity to schools for convenient access to education facilities and avoidance of proximity to the city's main

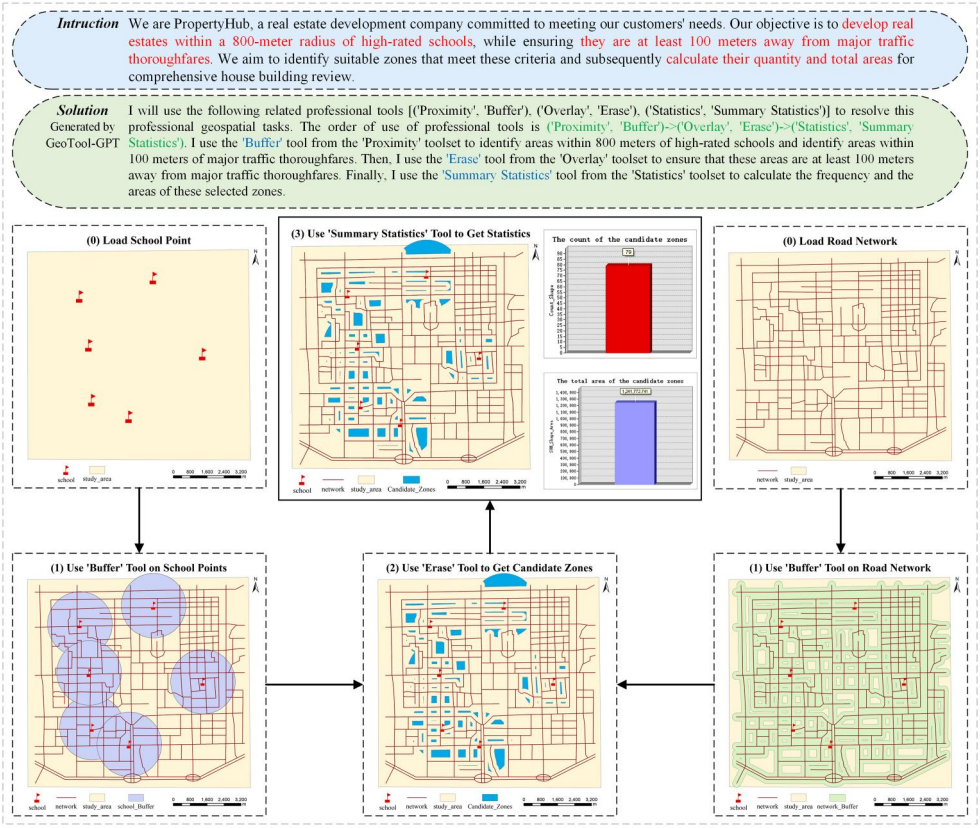

Figure 7. A case study demonstrating the use of three GIS tools to solve a simple GIS requirement involving spatial overlay analysis. The instruction provided within the blue block was input into GeoTool-GPT, yielding the model feedback presented in the green block. Then, the case study was implemented on the GIS software platform according to the specified tool sequence. The results after using the corresponding tools are displayed sequentially.

roads to minimize noise pollution. Subsequently, the identified areas necessitate statistical data such as quantity and area for further project evaluation.

We input the user's instructions into GeoTool-GPT, and received a solution, including the utilization of GIS tools and their sequence. Following the solution, the *Buffer* tools were applied to the spatial data of school and the road network, respectively. Subsequently, the *Erase* tool was employed to extract candidate zones that meet basic housing demand criteria. The *Summary Statistics* was finally utilized to obtain required statistics, such as the quantity and the total area of the candidate zones. The model's solution was then implemented on the collected data and the results are depicted in Figure 7. These results fulfilled the demand of the user and demonstrated the effectiveness of our GeoTool-GPT in responding to the instruction for a simple GIS requirement.

Second, we present an instruction for a relatively complex GIS requirement that involves using four GIS tool calls to solve the problem, as shown in Figure 8. Specifically, users need to locate the nearest landfill around the company and count

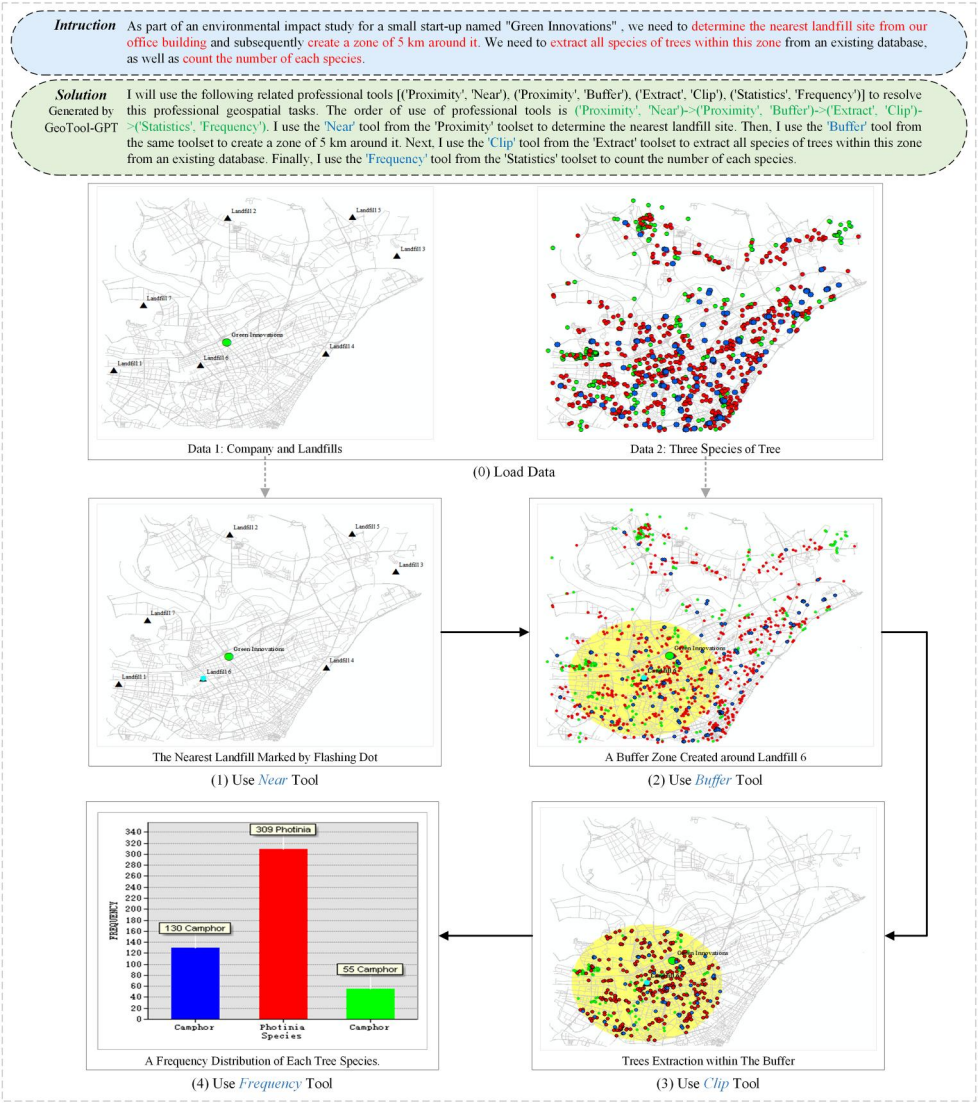

Figure 8. A case study demonstrating the use of four GIS tools to solve a relatively complex GIS requirement. The instruction provided within the blue block was input into GeoTool-GPT, yielding the model feedback presented in the green block. Then, the case study was implemented on the GIS software platform according to the specified tool sequence. The results after using the corresponding tools are displayed sequentially.

the number of tree species within a five-kilometer radius of it. Three datasets required to complete the instruction were loaded first, namely the point data of the company, the point data of its surrounding landfill, as well as the point data of each tree species. Following the solution given by the GeoTool-GPT, the *Near* tool was used first to get the nearest landfill from the company, and then the *Buffer* tool was used for creating a buffer zone around the nearest landfill. Subsequently, the *Clip* tool was used for

trees extraction within the buffer. Finally, the *Frequency* tool was used to count the species and numbers of the extracted tree data, in which the results obtained meet the instruction.

6. Discussion

6.1. Hierarchical organization of tools

In the field of GIS, the hierarchical organization of tools is a common practice, which is beneficial for enhancing tool positioning and usage efficiency among the numerous GIS tools available. In our study, it is similarly essential to organize the tools hierarchically within the geoprocessing toolkit. On one hand, as the number of GIS tools increases, it becomes challenging for GPT-4 to self-guide and generate instructions along with their corresponding solutions or responses based on all the tool documentation, due to the model's limitations imposed by its finite length of context window. On the other hand, to generate effective GeoSolution, GPT-4 needs to explore subsets of tools that are potentially relevant in usage, necessitating the potential classification of similar tools within the toolkit. An unorganized toolkit would increase the difficulty for the model to explore relevant tool subsets. Therefore, we advocate for a hierarchically organized division of the collected tools.

In this paper, organization of these tools is conducted based on their primary functionalities in handling GIS data. Specifically, we concentrate on two common GIS analysis functionalities: general analysis and advanced spatial analysis. Tools predominantly designed for general processing are integrated into the *Analysis* toolbox, while those specialized in advanced manipulation are integrated into in the *Spatial Analyst* toolbox. Drawing on domain expertise, multiple toolsets are curated for each toolbox, ensuring that tools in each toolset exhibit discernible similarities and interconnections in their usage. As an illustrative example, the *Overlay* toolset in the *Analysis* toolbox encompasses essential operators like erase, intersect, and union, which are tailored for overlaying multiple feature classes. Nevertheless, it is worth considering organizing the tools hierarchically from different perspectives to help the large language model better explore functionally related tools, thereby enhancing the richness of the data samples.

6.2. Limitations and scalability

Although the trained GeoTool-GPT has reached a level close to that of GPT-4 in solving some instructions, it was found in the evaluation process that its understanding of requirements and ability to use tools are limited when facing some complex GIS requirements. These limitations may be attributed to the quantity and quality of the training data. Enhancing the diversity of the data, particularly by constructing samples involving complex GIS task instructions and intricate sequences of GIS tool invocations, is a potentially effective method for addressing increasingly complex GIS tasks. This might require further refinement of the Self-Instruct strategy or direct improvements to system prompts, such as embedding complex case prompts within the system to help the model generate high-quality sample data. On the other hand, the training

data is limited by the cost of automatically generating samples based on GPT-4. Constructing more sample data to train superior or larger parameter open-source foundational models (e.g. LLaMA-3) is also an effective means of expanding the model's capabilities.

Additionally, GIS task requirements are diverse, and the GIS tools collected for this study may not cover the full spectrum of GIS needs, such as cartographic requirements. Therefore, the collection and organization of a broader range of GIS tools, following certain guidelines, are crucial for extending the model's ability to handle a wider array of geospatial tasks. Specifically, to expand capabilities for cartographic needs, developers could gather necessary cartographic sub-tools (e.g. adding legends, inserting submap), construct sample data using the method outlined in Section 3, and train a professional LLM based on the robust foundational model using the approach described in Section 4, thus customizing a specialized cartographic model. Integrating this model into an AI agent and linking it to relevant cartographic platforms would enable the automated resolution of cartographic tasks. Therefore, while the tools collected in this study are limited and focused on specific spatial analysis tasks rather than a broad spectrum, the method presented here is readily extendable to more extensive and complex geospatial tasks.

6.3. Application of GeoTool-GPT

The potential application scenarios for the trained GeoTool-GPT are primarily within AI Agent systems (Bran *et al*. 2023, Zhang *et al.* 2024a). An AI Agent, as a system capable of autonomously executing tasks, encompasses key functions such as perception, decision-making, and action (Xi *et al*. 2023). By positioning GeoTool-GPT as the core of the decision-making module, the system can perceive user needs through natural language on one hand, and on the other, provide solutions as a basis for action, thereby enabling the intelligent execution of GIS tasks.

Specifically, traditional GIS operations typically require users to switch between multiple tools and manually execute complex steps. By embedding GeoTool-GPT into an AI Agent, users need only describe their task requirements in natural language, and the AI Agent can understand the instructions by leveraging the language understanding capabilities of the augmented model, automatically dispatching the appropriate GIS tools to execute these tasks based on the solutions generated by the model. For instance, if a user requests 'Calculate the total area of all public green spaces in the city', the AI Agent can identify the steps involved in this task, such as data extraction, vector analysis, and statistical computation, and automatically invoke the relevant GIS tools to complete it. This combined capability of intelligent demand understanding and smart tool scheduling makes the resolution of GIS tasks more convenient and user-friendly.

7. Conclusion

In this paper, we focus on enhancing the GIS tool-use capabilities of LLMs. Specifically, we fine-tune GeoTool-GPT on the top of LLaMA-2-7B, a general-domain LLM, to equip it with specialized GIS tools knowledge for solving geospatial tasks using GIS tools.

To support the training process, we curated a comprehensive set of professional training data, including instruction-response samples (GeoTool) to adjust the general model's understanding of GIS tools, as well as instruction-solution samples (GeoSolution) to enhance the tool-use capability of the model. Additionally, we developed a benchmark dataset (GeoTask) for evaluating the GIS tool-use capabilities of LLMs. This dataset consists of GIS task instructions created and annotated by professional personnel to ensure reliable evaluation metrics. After the training process, GeoTool-GPT demonstrates robust GIS tool-use ability, consistently generating correct solutions for various geospatial tasks.

With this capability, GeoTool-GPT highlights its potential to automate complex geospatial workflows and significantly improve work efficiency. Moreover, despite the availability of numerous GIS tools in desktop software designed to meet diverse spatial analysis needs, their complex hierarchical structure and the gap between users' spatial reasoning and system implementation often hinder effective utilization, especially for non-expert users. GeoTool-GPT has the potential to bridge this gap, assisting non-professional users in navigating and leveraging GIS tools more effectively. Finally, the framework developed for GeoTool-GPT offers a practical approach for enhancing the tool-use capabilities of LLMs, which can serve as a reference for relevant researchers to develop corresponding models. Our curated training and evaluation datasets can also serve as valuable resources for researchers seeking to advance LLM applications in the GIS domain, fostering further innovation in this field.

However, there are limitations to this work. The current model, based on a 7 billion parameter base, may not fully capture the complexity of high-end GIS tasks, and the training data, while comprehensive, does not encompass the full range of GIS tools and scenarios. Future research could explore fine-tuning with even more diverse and granular task datasets, as well as investigating the scalability of GeoTool-GPT to larger parameter models. Additionally, integrating real-time data and multi-modal inputs (such as images and maps) could further enhance its applicability in dynamic GIS environments. This research lays the groundwork for future advancements in the integration of LLMs with GIS, presenting a promising direction for both academic and industrial applications.

Acknowledgments

The authors extend their sincere gratitude to the anonymous reviewers and the editor for their time, expertise, and constructive comments. Their insights and feedback have significantly enhanced the content of this article.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The project was supported by the National Natural Science Foundation of China [42371446 and 42071442], Natural Science Foundation of Hubei Province, China [2024AFD412], the

Fundamental Research Funds for the Central Universities, China, China University of Geosciences (Wuhan) [No. CUG170640], and by the Fundamental Research Funds for National Universities, China University of Geosciences [No. 2024XLA17]. This research was also supported by Meituan. Yifan Zhang acknowledges the support of the China Scholarship Council program [Project ID: 202306410085].

Notes on contributors

Cheng Wei is a master student in the School of Geography and Information Engineering, China University of Geosciences, Wuhan, China (CUG). His research interests include GeoAI and natural language processing techniques. His contribution includes conceptualization, methodology, data curation, software, visualization, writing – original draft, writing – review and editing, validation and supervision.

Yifan Zhang is a Ph.D. student in the School of Geography and Information Engineering China University of Geosciences, Wuhan, China (CUG). His research interests include GeoAI and map generalization. His contribution includes methodology, data curation, software, writing – original draft, writing – review and editing and supervision.

Xinru Zhao is a master student in the School of Geography and Information Engineering, China University of Geosciences, Wuhan, China (CUG). Her research interests include GeoAI and deep learning. Her contribution includes data curation and validation.

Ziyi Zeng is a Ph.D. student in the School of Geography and Information Engineering China University of Geosciences, Wuhan, China (CUG). Her research interests include GeoAI and LLMs. Her contribution includes data curation and validation.

Zhiyun Wang is an undergraduate student in the School of Geography and Information Engineering, China University of Geosciences, Wuhan, China (CUG). Her research interests include GeoAI and deep learning. Her contribution includes data curation and validation.

Jianfeng Lin is the leader of the cycling data mining team at Meituan, Beijing, China. His research interests include artificial intelligence and multimodal learning. His contribution includes supervision and project administration.

Qingfeng Guan is a professor at China University of Geosciences, Wuhan, China (CUG). His research interests include high-performance spatial computing, spatial computational intelligence, spatiotemporal big data. His contribution includes supervision and project administration.

Wenhao Yu received the B.S. and Ph.D. degrees in Geoinformatics from the Wuhan University, Wuhan, China, in 2010 and 2015, respectively. He is a professor at China University of Geosciences, Wuhan, China (CUG). His research interests include spatial data mining, map generalization, and LLMs. His contribution includes conceptualization, writing – original draft, writing – review and editing, supervision, validation, project administration and funding acquisition.

ORCID

Yifan Zhang http://orcid.org/0000-0002-5328-0881

Data and codes availability statement

The data and codes that support the findings of this study are available in [figshare.com] with the identifiers (https://doi.org/10.6084/m9.figshare.25486414).

References

Andrew, S., *et al.*, 2021. Habitat 2.0: training home assistants to rearrange their habitat. *In*: Advances in neural information processing systems (NeurIPS). arXiv:2106.14405.

Almarie, B., and Teixeira, P.E.P., 2023. The use of large language models in science: opportunities and challenges. *Theory and Practice of Digital Libraries*, 9 (1), 1.

Bran, A.M., *et al.*, 2023. ChemCrow: augmenting large-language models with chemistry tools. arXiv preprint arXiv:2304.05376.

Chang, Y., *et al.*, 2023. A survey on evaluation of large language models. arXiv preprint arXiv: 2307.03109.

Chiang, W., *et al.*, 2023. Vicuna: an open-source chatbot impressing GPT-4 with 90%* ChatGPT quality. https://vicuna.lmsys.org [accessed 14 April 2023].

Dai, H., *et al.*, 2023. AD-AutoGPT: an autonomous GPT for Alzheimer's disease infodemiology. arXiv preprint arXiv:2306.10095.

Deng, C., *et al.*, 2023. Learning a foundation language model for geoscience knowledge understanding and utilization. arXiv preprint arXiv:2306.05064.

Dídac, S., *et al.*, 2023. Vipergpt: visual inference via python execution for reasoning. arXiv preprint arXiv:2303.08128.

Ding, N., *et al.*, 2023. Enhancing chat language models by scaling high-quality instructional conversations. arXiv preprint arXiv:2305.14233.

Franceschelli, G., and Musolesi, M., 2023. On the creativity of large language models. arXiv preprint arXiv:2304.00008.

Guo, H., *et al.*, 2024. Remote sensing ChatGPT: solving remote sensing tasks with ChatGPT and visual models. arXiv preprint arXiv:2401.09083.

Gao, S., and Goodchild, M.F., 2013. Asking spatial questions to identify GIS functionality. In: 2013 *Fourth international conference on computing for geospatial research and application*, 106–110. IEEE.

Gao, Y., *et al.*, 2023. Chat-rec: towards interactive and explainable LLMS-augmented recommender system. arXiv preprint arXiv:2303.14524.

Hadi, M.U., *et al.*, 2023. *A survey on large language models: applications, challenges, limitations, and practical usage*. TechRxiv. https://doi.org/10.36227/techrxiv.23589741.v1

Hao, W., *et al.*, 2020. Towards learning a generic agent for vision-and-language navigation via pre-training. *In*: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 13137–13146.

Hu, E.J., *et al.*, 2021. Lora: low-rank adaptation of large language models. arXiv preprint arXiv: 2106.09685.

Hochmair, H.H., and Guan, B., 2023. ChatGPT as a mapping assistant: a novel method to enrich maps with generative AI and content derived from street-level photographs.

Hu, Y., *et al.*, 2023. Geo-knowledge-guided GPT models improve the extraction of location descriptions from disaster-related social media messages. *International Journal of Geographical Information Science*, 37 (11), 2289–2318.

Iyer, C.V., K., Ganguli, S., and Pandey, V., 2023. Perspectives on geospatial artificial intelligence platforms for multimodal spatiotemporal datasets. *In*: C. V. Krishnakumar Iyer, Swetava Ganguli, and Vipul Pandey, eds. Advances in *scalable and intelligent geospatial analytics*. Boca Raton, FL: CRC Press, 17–63.

Janowicz, K., *et al.*, 2020. GeoAI: spatially explicit artificial intelligence techniques for geographic knowledge discovery and beyond. *International Journal of Geographical Information Science*, 34 (4), 625–636.

Jin, K., and Zhuo, H.H., 2022. Integrating AI planning with natural language processing: a combination of explicit and tacit knowledge. arXiv preprint arXiv:2202.07138.

Ji, Y., and Gao, S., 2023. Evaluating the effectiveness of large language models in representing textual descriptions of geometry and spatial relations. arXiv preprint arXiv:2307.03678.

Karabacak, M., and Margetis, K., 2023. Embracing large language models for medical applications: opportunities and challenges. *Cureus*, 15 (5), e39305.

Li, Z., and Ning, H., 2023. Autonomous GIS: the next-generation AI-powered GIS. arXiv preprint arXiv:2305.06453.

Li, C., *et al.*, 2023. Multimodal foundation models: from specialists to general-purpose assistants. arXiv preprint arXiv:2309.10020.

Li, C., *et al.*, 2022. ELEVATER: a benchmark and toolkit for evaluating language-augmented visual models. *In*: *NeurIPS track on datasets and benchmarks*. arXiv:2204.08790.

Li, W., 2020. GeoAI: where machine learning and big data converge in GIScience. *Journal of Spatial Information Science*, 20, 71–77.

Li, J., *et al.*, 2022b. Pretrained language models for text generation: a survey. arXiv preprint arXiv:2201.05273.

Mai, G., *et al.*, 2023. On the opportunities and challenges of foundation models for geospatial artificial intelligence. arXiv preprint arXiv:2304.06798.

Mai, G., *et al.*, 2022. A review of location encoding for GeoAI: methods and applications. *International Journal of Geographical Information Science*, 36 (4), 639–673.

OpenAI, 2023a., ChatGPT. https://openai.com/blog/chatgpt/

OpenAI, 2023b., GPT-4 technical report. arXiv preprint arXiv:2303.08774.

Qin, Y., *et al.*, 2023. Toolllm: facilitating large language models to master 16000+ real-world apis. arXiv preprint arXiv:2307.16789.

Rae, J.W., *et al.*, 2021. Scaling language models: methods, analysis & insights from training gopher. arXiv preprint arXiv:2112.11446.

Roberts, J., *et al.*, 2023. GPT4GEO: how a language model sees the world's geography. arXiv preprint arXiv:2306.00020.

Rogers, A., Gardner, M., and Augenstein, I., 2023. QA dataset explosion: a taxonomy of NLP resources for question answering and reading comprehension. *ACM Computing Surveys*, 55 (10), 1–45.

Shen, W., *et al.*, 2024. Small LLMS are weak tool learners: a multi-LLM agent. arXiv preprint arXiv: 2401.07324.

Sun, Y., *et al.*, 2021. Ernie 3.0: large-scale knowledge enhanced pre-training for language understanding and generation. arXiv preprint arXiv:2107.02137.

Tanmay, G., and Aniruddha, K., 2022. Visual programming: compositional visual reasoning without training. arXiv preprint arXiv:2211.11559.

Taori, R., *et al.*, 2023. Alpaca: a strong, replicable instruction-following model. *Stanford Center for Research on Foundation Models*, 3 (6), 7. https://crfm.stanford.edu/2023/03/13/alpaca.html,

Teubner, T., *et al.*, 2023. Welcome to the era of ChatGPT et al. the prospects of large language models. *Business & Information Systems Engineering*, 65 (2), 95–101.

Thirunavukarasu, A.J., *et al.*, 2023. Large language models in medicine. *Nature Medicine*, 29 (8), 1930–1940.

Tim, B., *et al.*, 2022. Instruct pix2pix: learning to follow image editing instructions. arXiv preprint arXiv:2211.09800.

Touvron, H., *et al.*, 2023. Llama 2: open foundation and fine-tuned chat models. arXiv preprint arXiv:2307.09288.

Tang, Q., *et al.*, 2023. Toolalpaca: generalized tool learning for language models with 3000 simulated cases. arXiv preprint arXiv:2306.05301.

Tao, R., and Xu, J., 2023. Mapping with ChatGPT. *ISPRS International Journal of Geo-Information*, 12 (7), 284.

Wang, Y., *et al.*, 2022. Self-instruct: aligning language model with self generated instructions. arXiv preprint arXiv:2212.10560.

Wei, J., *et al.*, 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35, 24824–24837.

Wu, C., *et al.*, 2023. Visual chatgpt: talking, drawing and editing with visual foundation models. arXiv preprint arXiv:2303.04671.

Xu, C., *et al.*, 2023. Baize: an open-source chat model with parameter-efficient tuning on self-chat data. arXiv preprint arXiv:2304.01196.

Xi, Z., *et al.*, 2023. The rise and potential of large language model based agents: a survey. arXiv preprint arXiv:2309.07864.

Yang, Z., *et al.*, 2023. Mm-react: prompting ChatGPT for multimodal reasoning and action. arXiv preprint arXiv:2303.11381.

Yue, S., *et al.*, 2023. DISC-LawLLM: fine-tuning large language models for intelligent legal services. arXiv preprint arXiv:2309.11325.

Zhang, R., *et al.*, 2023. Prompt, generate, then cache: cascade of foundation models makes strong few-shot learners. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognitio*n, 15211–15222.

Zhan, Y., Xiong, Z., and Yuan, Y., 2024. SkyEyeGPT: unifying remote sensing vision-language tasks via instruction tuning with large language model. arXiv preprint arXiv:2401.09712.

Zhang, Y., *et al.*, 2024a. GeoGPT: an assistant for understanding and processing geospatial tasks. *International Journal of Applied Earth Observation and Geoinformation*, 131, 103976.

Zhang, Y., *et al.*, 2024b. MapGPT: an autonomous framework for mapping by integrating large language model and cartographic tools. *Cartography and Geographic Information Science*, 51 (6), 717–743. https://doi.org/10.1080/15230406.2024.2404868

Zhang, Y., *et al.*, 2024c. BB-GeoGPT: A framework for learning a large language model for geographic information science. *Information Processing & Management*, 61 (5), 103808.

Zhu, A.X., *et al.*, 2021. Next generation of GIS: must be easy. *Annals of GIS*, 27 (1), 71–86.

Zou, X., *et al.*, 2022. Generalized decoding for pixel, image, and language. arXiv preprint arXiv: 2212.11270.

Appendix A

Table A1. The hierarchical organization of GIS tools for LLM agents.

Toolbox name	Toolset name	Tool name
Analysis (primarily designed for general analysis)	Extract	Clip, Select, Split, Table Select
	Overlay	Erase, Identity, Intersect, Spatial Join, Symmetrical Difference, Union, Update
	Proximity	Buffer, Create Thiessen Polygons, Generate Near Table, Multiple Ring Buffer, Near, Point Distance, Polygon Neighbors
	Statistics	Frequency, Summary Statistics, Tabulate Intersection
Spatial Analyst (primarily designed for advanced spatial analysis)	Conditional	Con, Pick, Set Null
	Density	Kernel Density, Line Density, Point Density
	Distance	Corridor, Cost Allocation, Cost Back Link, Cost Distance, Cost Path, Euclidean Allocation, Euclidean Direction, Euclidean Distance, Path Distance, Path Distance Allocation, Path Distance Back Link
	Extraction	Extract by Attributes, Extract by Circle, Extract by Mask, Extract by Points, Extract by Polygon, Extract by Rectangle, Extract Multi Values to Points, Extract Values to Points, Sample
	Generalization	Aggregate, Boundary Clean, Expand, Majority Filter, Nibble, Region Group, Shrink, Thin
	Groundwater	Darcy Flow, Darcy Velocity, Particle Track, Porous Puff
	Hydrology	Basin, Fill, Flow Accumulation, Flow Direction, Flow Length, Sink, Snap Pour Point, Stream Link, Stream Order, Stream to Feature, Watershed
	Interpolation	IDW, Kriging, Natural Neighbor, Spline, Spline with Barriers, Topo to Raster, Topo to Raster by File, Trend
	Local	Cell Statistics, Combine, Equal To Frequency, Greater Than Frequency, Highest Position, Less Than Frequency, Lowest Position, Popularity, Rank
	Map Algebra	Raster Calculator
	General Math	Abs, Divide, Exp, Exp10, Exp2, Float, Int, Ln, Log10, Log2, Minus, Mod, Negate, Plus, Power, Round Down, Round Up, Square, Square Root, Times
	Logical Math	Boolean And, Boolean Not, Boolean Or, Boolean XOr, Combinatorial And, Combinatorial Or, Combinatorial XOr, Equal To, Greater Than, Greater Than Equal, Less Than, Less Than Equal, Not Equal, Diff, InList, Is Null, Over, Test
	Multivariate	Band Collection Statistics, Class Probability, Create Signatures, Dendrogram, Edit Signatures, Iso Cluster, Iso Cluster Unsupervised Classification, Maximum Likelihood Classification, Principal Components
	Neighborhood	Block Statistics, Filter, Focal Flow, Focal Statistics, Line Statistics, Point Statistics
	Overlay Analysis	Fuzzy Membership, Fuzzy Overlay, Weighted Overlay, Weighted Sum
	Raster Creation	Create Constant Raster, Create Normal Raster, Create Random Raster
	Reclass	Lookup, Reclass by ASCII File, Reclass by Table, Reclassify, Rescale by Function, Slice
	Surface	Aspect, Contour, Contour List, Contour with Barriers, Curvature, Cut Fill, Hillshade, Observer Points, Slope, Viewshed, Visibility
	Zonal	Tabulate Area, Zonal Fill, Zonal Geometry, Zonal Geometry As Table, Zonal Histogram, Zonal Statistics, Zonal Statistics as Table
Total	23	172